



Docker

This is a quick reference cheat sheet for Docker. And you can find the most common Docker commands here.

Getting Started

Getting started

Create and run a container in background

```
$ docker run -d -p 80:80 docker/getting-started
```

-d - Run the container in detached mode

-p 80:80 - Map port 80 to port 80 in the container

docker/getting-started - The image to use

Create and run a container in foreground

```
$ docker run -it -p 8001:8080 --name my-nginx nginx
```

-it - Interactive bash mode

-p 8001:8080 - Map port 8001 to port 8080 in the container

--name my-nginx - Specify a name

nginx - The image to use

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General commands



<code>docker ps</code>	List running containers
<code>docker ps -a</code>	List all containers
<code>docker ps -s</code>	List running containers (with CPU / memory)
<code>docker images</code>	List all images
<code>docker exec -it <container> bash</code>	Connecting to container
<code>docker logs <container></code>	Shows container's console log
<code>docker stop <container></code>	Stop a container
<code>docker restart <container></code>	Restart a container
<code>docker rm <container></code>	Remove a container
<code>docker port <container></code>	Shows container's port mapping
<code>docker top <container></code>	List processes
<code>docker kill <container></code>	Kill a container
Parameter <code><container></code> can be container id or name	

Docker Containers

Starting & Stopping

<code>docker start my-nginx</code>	Starting
<code>docker stop my-nginx</code>	Stopping
<code>docker restart my-nginx</code>	Restarting
<code>docker pause my-nginx</code>	Pausing
<code>docker unpause my-nginx</code>	Unpausing
<code>docker wait my-nginx</code>	Blocking a Container
<code>docker kill my-nginx</code>	Sending a SIGKILL

```
docker attach my-nginx
```

Connecting to an Existing Container

Information

<code>docker ps</code>	List running containers
<code>docker ps -a</code>	List all containers
<code>docker logs my-nginx</code>	Container Logs
<code>docker inspect my-nginx</code>	Inspecting Containers
<code>docker events my-nginx</code>	Containers Events
<code>docker port my-nginx</code>	Public Ports
<code>docker top my-nginx</code>	Running Processes
<code>docker stats my-nginx</code>	Container Resource Usage
<code>docker diff my-nginx</code>	Lists the changes made to a container.

Creating

```
docker create [options] IMAGE
-a, --attach                # attach stdout/err
-i, --interactive           # attach stdin (interactive)
-t, --tty                   # pseudo-tty
    --name NAME             # name your image
-p, --publish 5000:5000    # port map (host:container)
    --expose 5432           # expose a port to containers
-P, --publish-all         # publish all ports
    --link container:alias  # linking
-v, --volume `pwd`:/app    # mount (absolute paths needed)
-e, --env NAME=hello       # env vars
```

Example

```
$ docker create --name my_redis --expose 6379 redis:3.0.2
```

Manipulating

Renaming a Container

```
docker rename my-nginx my-nginx
```

Removing a Container

```
docker rm my-nginx
```

Updating a Container

```
docker update --cpu-shares 512 -m 300M my-nginx
```

Docker Images

Manipulating

<code>docker images</code>	Listing images
<code>docker rmi nginx</code>	Removing an image
<code>docker load < ubuntu.tar.gz</code>	Loading a tarred repository
<code>docker load --input ubuntu.tar</code>	Loading a tarred repository
<code>docker save busybox > ubuntu.tar</code>	Save an image to a tar archive
<code>docker history</code>	Showing the history of an image
<code>docker commit nginx</code>	Save a container as an image.
<code>docker tag nginx eon01/nginx</code>	Tagging an image
<code>docker push eon01/nginx</code>	Pushing an image

Building Images

```
$ docker build .
$ docker build github.com/creack/docker-firefox
$ docker build - < Dockerfile
$ docker build - < context.tar.gz
$ docker build -t eon/my-nginx .
$ docker build -f myOtherDockerfile .
$ curl example.com/remote/Dockerfile | docker build -f - .
```

Docker Networking

Manipulating

Removing a network

```
docker network rm MyOverlayNetwork
```

Listing networks

```
docker network ls
```

Getting information about a network

```
docker network inspect MyOverlayNetwork
```

Connecting a running container to a network

```
docker network connect MyOverlayNetwork nginx
```

Connecting a container to a network when it starts

```
docker run -it -d --network=MyOverlayNetwork nginx
```

Disconnecting a container from a network

```
docker network disconnect MyOverlayNetwork nginx
```

Creating Networks

```
docker network create -d overlay MyOverlayNetwork
```

```
docker network create -d bridge MyBridgeNetwork
```

```
docker network create -d overlay \  
  --subnet=192.168.0.0/16 \  
  --subnet=192.170.0.0/16 \  
  --gateway=192.168.0.100 \  
  --gateway=192.170.0.100 \  
  --ip-range=192.168.1.0/24 \  

```

```
--aux-address="my-router=192.168.1.5" \  
--aux-address="my-switch=192.168.1.6" \  
--aux-address="my-printer=192.170.1.5" \  
--aux-address="my-nas=192.170.1.6" \  
MyOverlayNetwork
```

Clean Up

Clean All

Cleans up dangling images, containers, volumes, and networks (ie, not associated with a container)

```
docker system prune
```

Additionally, remove any stopped containers and all unused images (not just dangling images)

```
docker system prune -a
```

Containers

Stop all running containers

```
docker stop $(docker ps -a -q)
```

Delete stopped containers

```
docker container prune
```

Images

Remove all dangling (not tagged and is not associated with a container) images:

```
docker image prune
```

Remove all images which are not used by existing containers

```
docker image prune -a
```

Volumes

```
docker volume prune
```

Remove all volumes not used by at least one container

Miscellaneous

Docker Hub

```
docker search search_word
```

Search docker hub for images.

```
docker pull user/image
```

Downloads an image from docker hub.

```
docker login
```

Authenticate to docker hub

```
docker push user/image
```

Uploads an image to docker hub.

Registry commands

Login to a Registry

```
$ docker login
$ docker login localhost:8080
```

Logout from a Registry

```
$ docker logout
$ docker logout localhost:8080
```

Searching an Image

```
$ docker search nginx
$ docker search nginx --stars=3 --no-trunc busybox
```

Pulling an Image

```
$ docker pull nginx
$ docker pull eon01/nginx localhost:5000/myadmin/nginx
```

Pushing an Image

```
$ docker push eon01/nginx
$ docker push eon01/nginx localhost:5000/myadmin/nginx
```

Batch clean

<code>docker stop -f \$(docker ps -a -q)</code>	Stopping all containers
<code>docker rm -f \$(docker ps -a -q)</code>	Removing all containers
<code>docker rmi -f \$(docker images -q)</code>	Removing all images

Volumes

Check volumes

```
$ docker volume ls
```

Cleanup unused volumes

```
$ docker volume prune
```

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